Team 3

Prompt 2: Geospatial AI Crop Yield Forecasting

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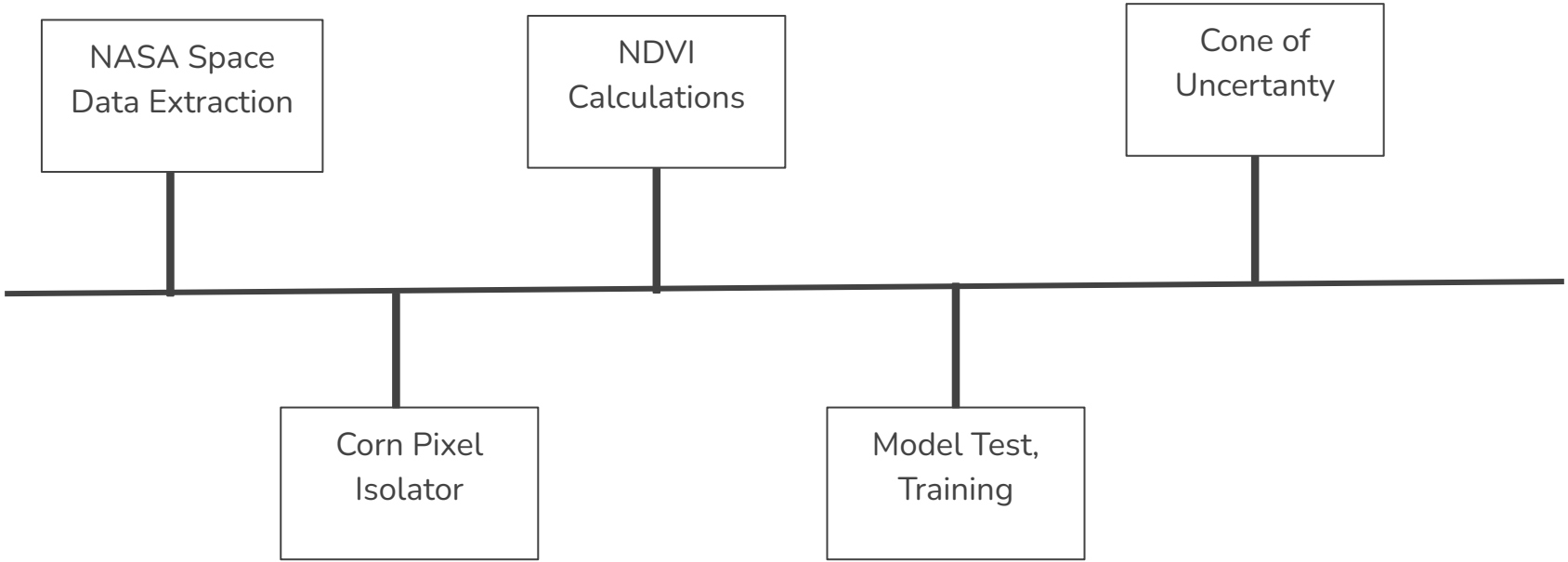
Our Data Pipeline



Extracting Data From Earth Data

The Hurdles:

- Interpreting the HLS.Tutorial file and Problem statement
- Working with AWS for the first time
- AWS/GitHub physical memory limit

The Pull-Through:

- Analysing the problem, only taking what was needed, dividing tasks
- Utilizing AWS and other tools and people for guidance
- Analyzing and cutting down excess data

```
QUEUEING TASKS | : 100%  300/300 [00:00<00:00, 14726.16it/s]
PROCESSING TASKS | : 100%  300/300 [00:20<00:00, 18.25it/s]
COLLECTING RESULTS | : 100%  300/300 [00:00<00:00, 56370.00it/s]
  Downloaded files to ./cornYield/Colorado/
Searching data for Wisconsin...
  Found 100 granules for Wisconsin
  Found 300 matching band files for Wisconsin
QUEUEING TASKS | : 100%  300/300 [00:00<00:00, 16718.15it/s]
PROCESSING TASKS | : 100%  300/300 [00:21<00:00,  1.79it/s]
COLLECTING RESULTS | : 100%  300/300 [00:00<00:00, 52736.43it/s]
  Downloaded files to ./cornYield/Wisconsin/
Searching data for Missouri...
  Found 100 granules for Missouri
  Found 300 matching band files for Missouri
QUEUEING TASKS | : 100%  300/300 [00:00<00:00, 17559.92it/s]
PROCESSING TASKS | : 100%  300/300 [00:16<00:00, 17.02it/s]
COLLECTING RESULTS | : 100%  300/300 [00:00<00:00, 52448.47it/s]
  Downloaded files to ./cornYield/Missouri/
Searching data for Nebraska...
  Found 100 granules for Nebraska
  Found 300 matching band files for Nebraska
QUEUEING TASKS | : 100%  300/300 [00:00<00:00, 15372.95it/s]
PROCESSING TASKS | : 100%  300/300 [00:15<00:00, 24.91it/s]
COLLECTING RESULTS | : 100%  300/300 [00:00<00:00, 60142.01it/s]
  Downloaded files to ./cornYield/Nebraska/
```

Finding Corn from .tif files

1. Use rasterio
2. USDA Cropland Data Layer(CDL)
3. Corn mask
4. Save only files with corn
5. Count corn pixels(optional)

```
with rasterio.open(e.path) as src:
        data = src.read(1)
        profile = src.profile

corn_mask = np.where(data == 1, 255, 0).astype(np.uint8)
```

Cropland Data Layer (CDL) Key

The Cropland Data Layer (CDL) fields are populated by Regrid from the USDA's nationwide raster dataset. They have an extensive and detailed, highly technical FAQ and research references available.

CDL data is only available on parcels 12 acres or larger.

Regrid attributes are derived from the 2022 CDL, released February 2023.

The acreage numbers listed below are directly from the USDA's nationwide summary information. Values with no category and/or zero (0) acreage are not currently used by the USDA and will not be in our derived attributes.

Value	Category	Acreage
1	Corn	93,391,174
2	Cotton	13,683,288
3	Rice	3,164,624
4	Sorghum	7,107,474
5	Soybeans	84,569,951
6	Sunflower	1,563,003
7		0
8		0
9		0
10	Peanuts	1,727,638
11	Tobacco	103,467
12	Sweet Corn	281,995
13	Pop or Orn Corn	187,344
14	Mint	77,746

```
Generating Bedrock predictions...
Bedrock error: An error occurred (AccessDeniedException) when calling the InvokeModel operation: Model access is denied due
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 try again after 2 minutes.
```

Vibe coding Bedrock AI

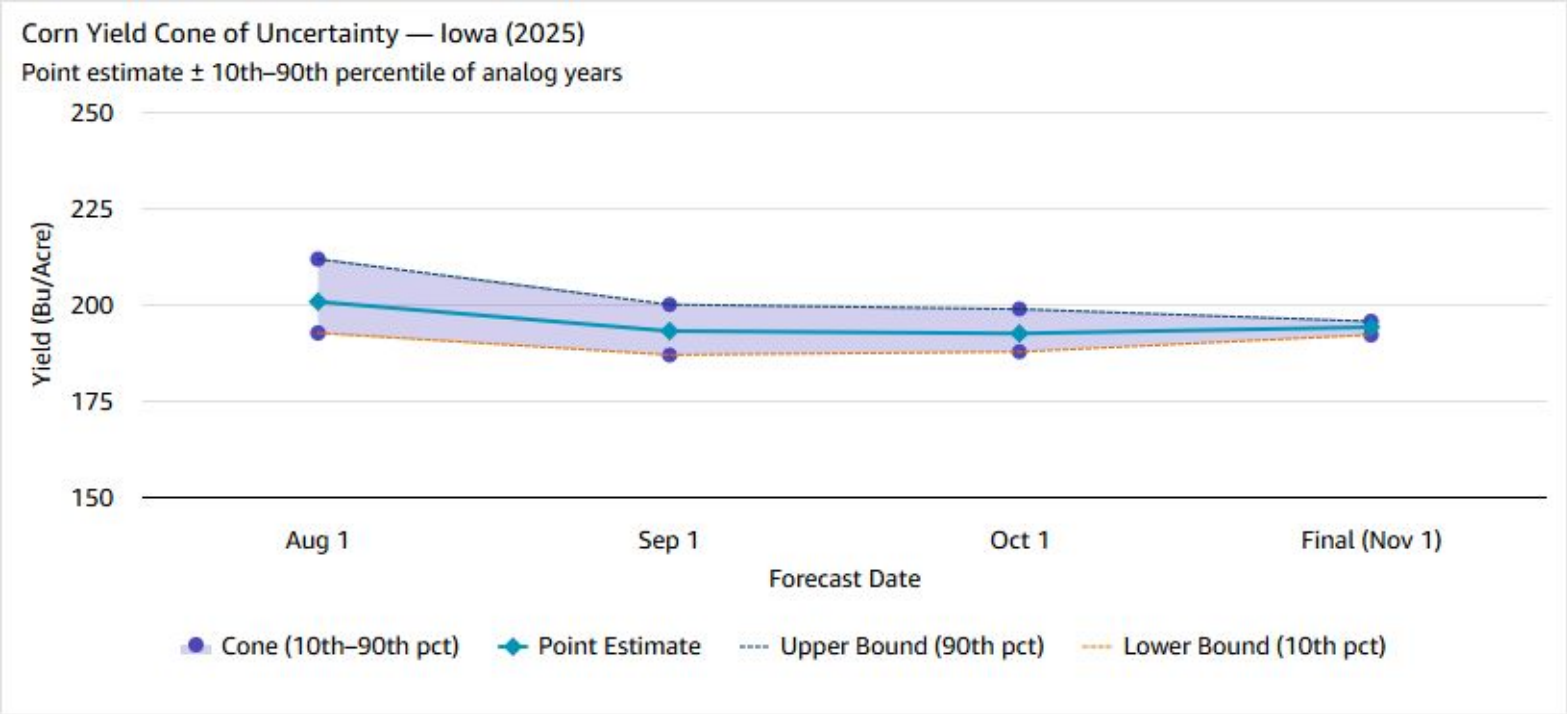
I thought making an AI model would be very easy by just letting Amazon Q configure Bedrock for me. I was wrong.

```
Bedrock Corn Yield Predictions 2025:
============================================================
    state forecast_date  point_estimate  lower_10pct  upper_90pct  cone_width
 Colorado          Aug1           158.0        134.3        181.7        47.4
 Colorado          Sep1           158.0        139.0        176.9        37.9
 Colorado          Oct1           158.0        145.3        170.6        25.3
 Colorado         Final           158.0        150.1        165.9        15.8
     Iowa          Aug1           184.2        156.6        211.9        55.3
     Iowa          Sep1           184.2        162.1        206.3        44.2
     Iowa          Oct1           184.2        169.5        199.0        29.5
     Iowa         Final           184.2        175.0        193.4        18.4
 Missouri          Aug1           159.7        135.7        183.7        48.0
 Missouri          Sep1           159.7        140.5        178.9        38.4
 Missouri          Oct1           159.7        146.9        172.5        25.6
 Missouri         Final           159.7        151.7        167.7        16.0
 Nebraska          Aug1           180.1        153.1        207.1        54.0
 Nebraska          Sep1           180.1        158.5        201.7        43.2
 Nebraska          Oct1           180.1        165.7        194.5        28.8
 Nebraska         Final           180.1        171.1        189.1        18.0
Wisconsin          Aug1           180.4        153.4        207.5        54.1
Wisconsin          Sep1           180.4        158.8        202.1        43.3
Wisconsin          Oct1           180.4        166.0        194.9        28.9
Wisconsin         Final           180.4        171.4        189.4        18.0

Results saved to: bedrock_corn_predictions_2025.csv

Summary by State:
          avg_yield  avg_uncertainty
state
Colorado      158.0             31.6
Iowa          184.2             36.8
Missouri      159.7             32.0
Nebraska      180.1             36.0
Wisconsin     180.4             36.1
✅Bedrock pipeline complete
sagemaker-user@default:~$
```

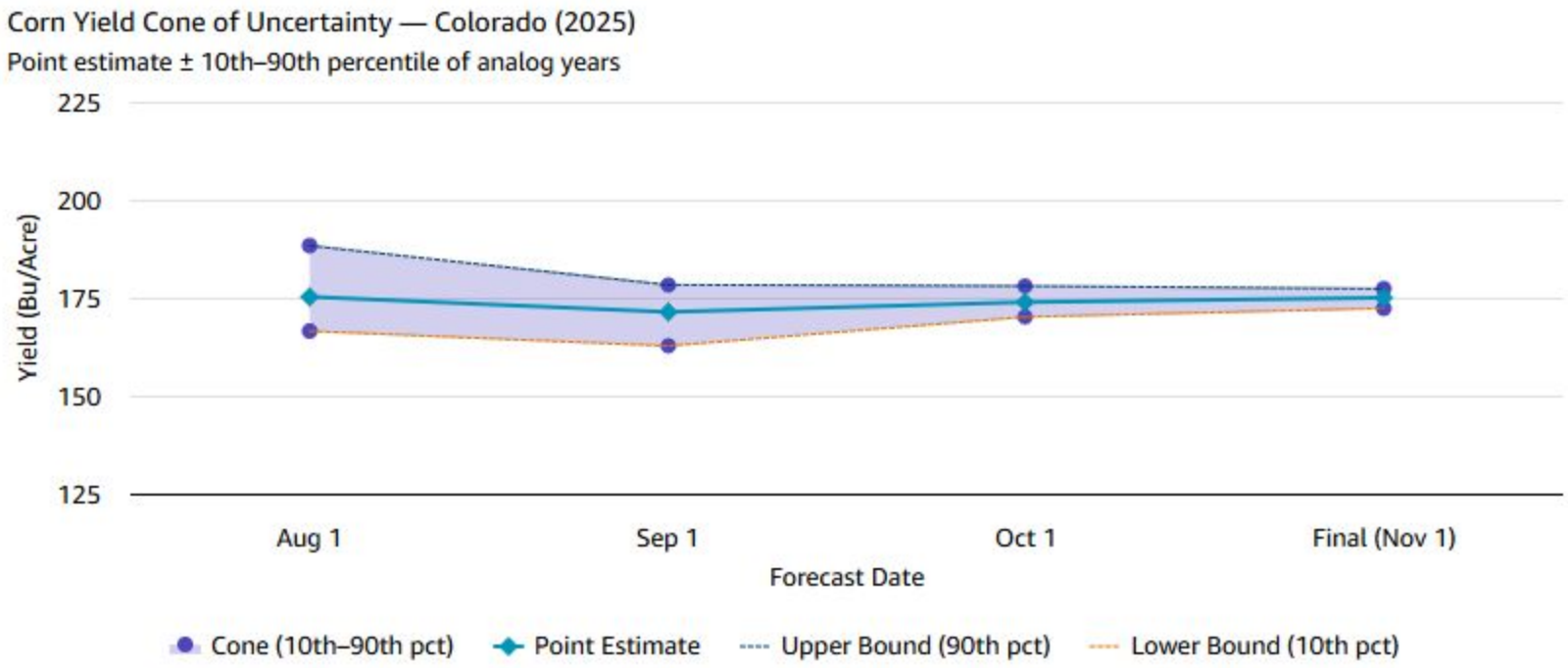
Cone of Uncertainty Process:

- Used AWS to generate a python script of the cone of uncertainty for simulated data and real data.
- Used the python script as a guide for AWS to create visuals of the cone of uncertainty for each state and forecast data using simulated data.

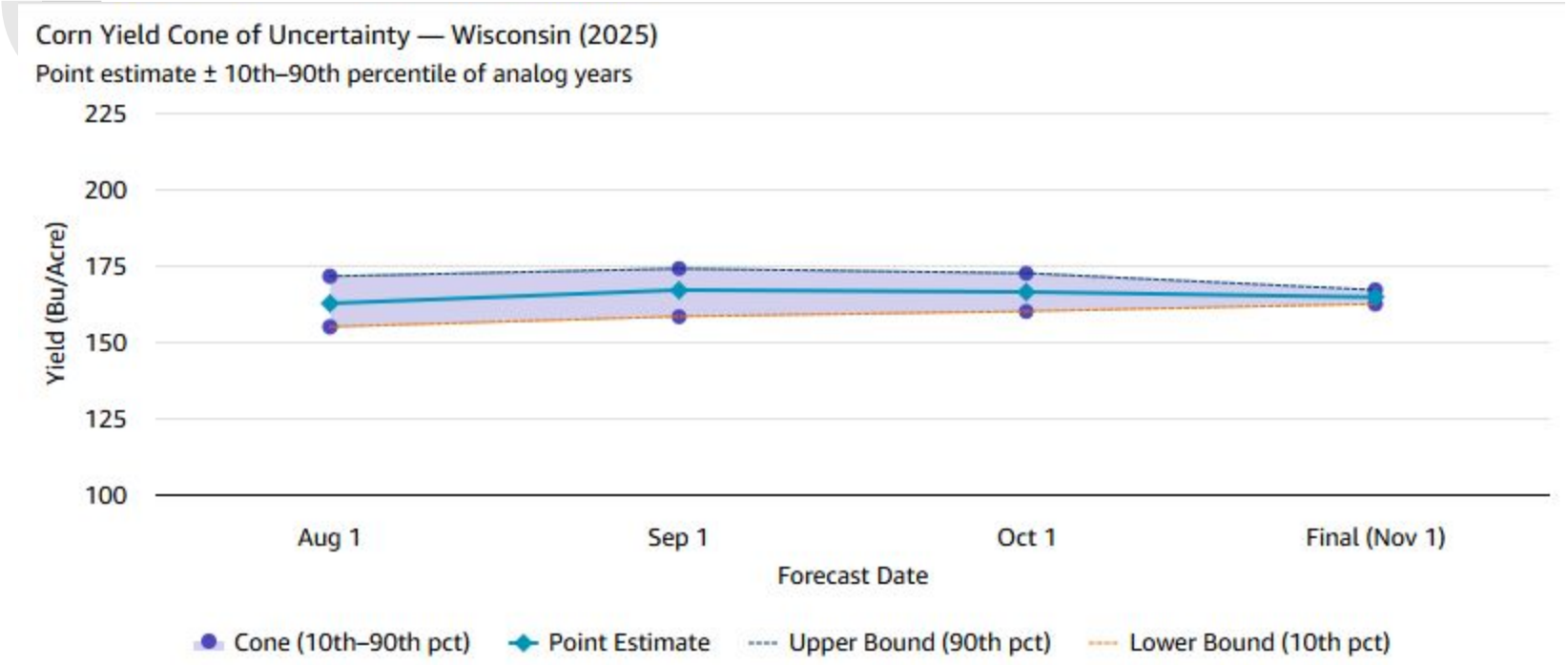
Cone of uncertainty: Iowa



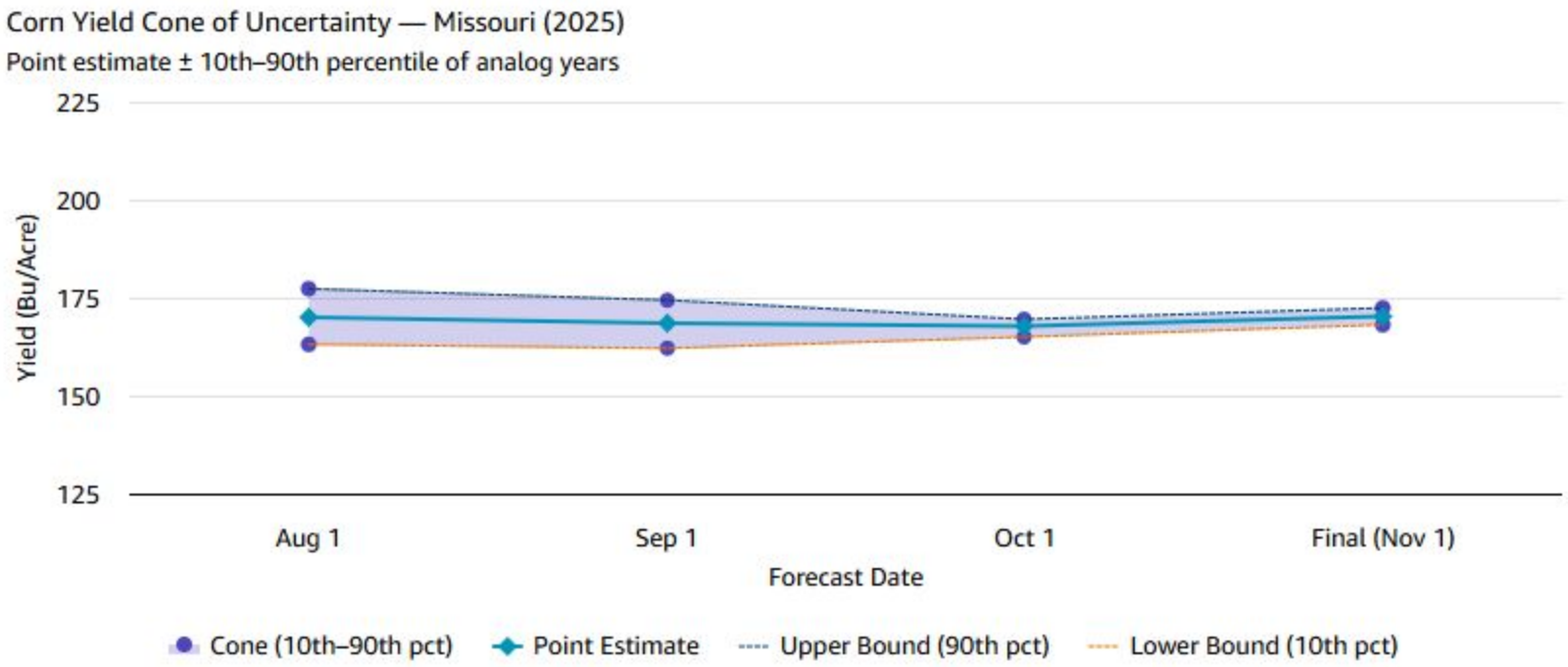
Cone of uncertainty: Colorado



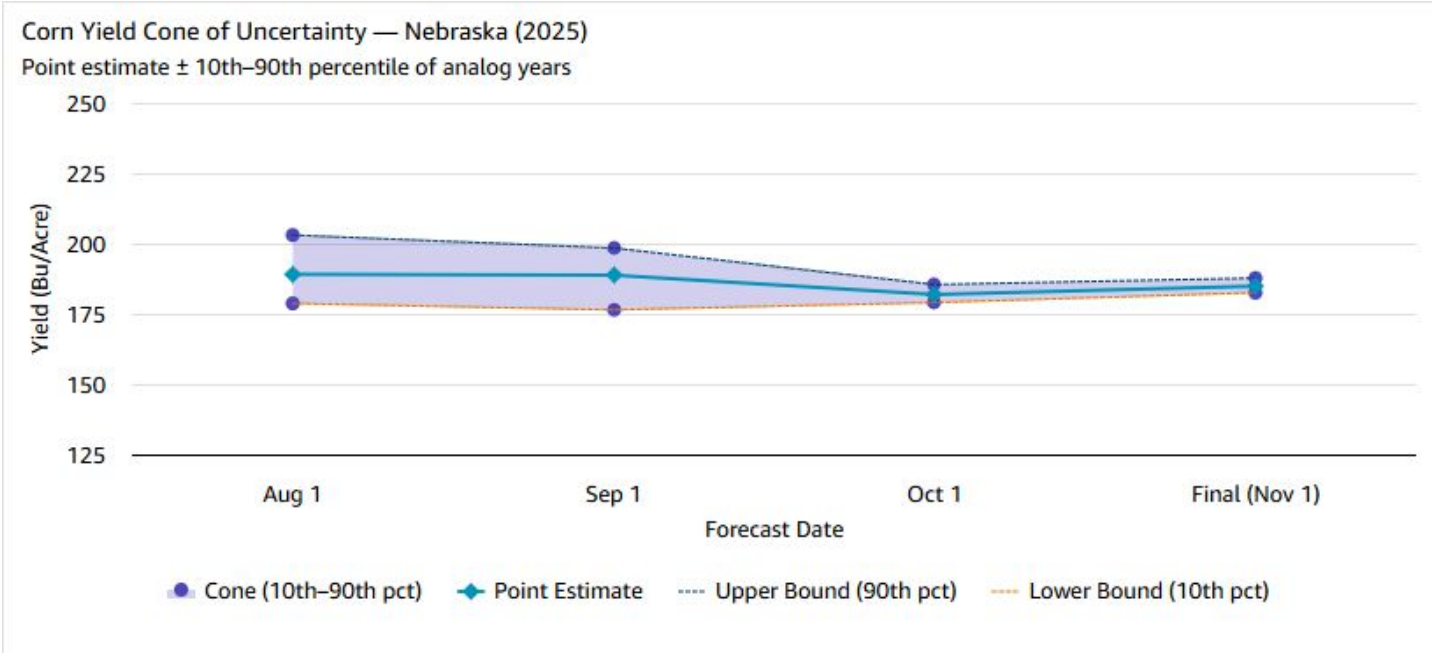
Cone of uncertainty: Wisconsin



Cone of uncertainty: Missouri



Cone of uncertainty: Nebraska



2025 Predictions:

STEP 7: Summary — Average Yield Forecast per State (2025)

State	Avg Yield (Bu/Ac)	Avg Lower	Avg Upper	Avg Cone Width
Colorado	149.6	140.9	160.1	19.2
Iowa	180.3	168.6	191.7	23.1
Missouri	149.3	129.6	166.9	37.3
Nebraska	173.0	162.7	183.0	20.2
Wisconsin	166.0	151.5	181.2	29.6